Computer Science Theory Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (20 marks)

1. Which of the following is correct about Python?
   (a) It supports automatic garbage collection.
   (b) It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
   (c) Both of the above.
   (d) None of the above.

2. Church's thesis equates the concept of "computable function" with those functions computable by, for example, Turing machines. Which of the following is true...
   (a) It was first proven by Alan Turing.
   (b) It has not yet been proven, but finding a proof is a subject of active research.
   (c) It can never be proven.
   (d) It is now in doubt because of the advent of parallel computers.

3. A large Java program was tested extensively, and no errors were found. What can be concluded?
   (a) All of the preconditions in the program are correct.
   (b) All of the postconditions in the program are correct.
   (c) The program may have bugs.
   (d) Every method in the program may safely be used in other programs.

4. Which of the following characteristics of a programming language is best specified using a context-free grammar?
   (a) Identifier length
   (b) Maximum level of nesting
   (c) Operator precedence
   (d) Type compatibility

5. Which of the following is NOT a property of bitmap graphics?
   (a) Fast hardware exists to move blocks of pixels efficiently.
   (b) Realistic lighting and shading can be done.
   (c) All line segments can be displayed as straight.
   (d) Polygons can be filled with solid colors and textures.

6. In Python 3, what is the output of ['Hi!'] * 4?

'Hi!', 'Hi!', 'Hi!', 'Hi!'

'Hi!' * 4

(a) Error

(b) None of the above.

7. Which of the following has the greatest potential for compromising a user's personal privacy?

   (a) A group of cookies stored by the user's Web browser
   (b) The Internet Protocol (IP) address of the user's computer
   (c) The user's e-mail address
   (d) The user's public key used for encryption

8. Which of the following pairs of 8-bit, two's-complement numbers will result in overflow when the members of the pairs are added?

   (a) 11111111, 00000001
   (b) 00000001, 10000000
   (c) 11111111, 10000001
   (d) 10000001, 10101010

9. In python 3, which of the following is floor division?

   (a) /
   (b) //
   (c) %
   (d) —

10. Let x = 8. What is x¿¿1 in Python 3?

   (a) 3
   (b) 4
   (c) 2
   (d) 8

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Of the following problems concerning a given undirected graph G, which is currently known to be solvable in polynomial time?' is 'Finding a shortest cycle in G'.

2. True or False: The correct answer to 'Local caching of files is common in distributed file systems, but it has the disadvantage that' is 'A much higher amount of network traffic results'.

3. True or False: The correct answer to 'Which of the following is true of interrupts?' is 'They are generated when memory cycles are "stolen".'.

4. True or False: The correct answer to 'Which is the smallest asymptotically?' is 'O(n)'.

5. True or False: The correct answer to 'Any set of Boolean operators that is sufficient to represent all Boolean expressions is said to be complete. Which of the following is...' is '{AND, OR}'.

6. True or False: The correct answer to 'In Python 3, what is the output of print tuple[0] if tuple = ('abcd', 786 , 2.23, 'john', 70.2)?' is 'abcd'.

7. True or False: The correct answer to 'Of the following sorting algorithms, which has a running time that is LEAST dependent on the initial ordering of the input?' is 'Merge sort'.

8. True or False: The correct answer to 'Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-...' is 'The number of computers launching the attack'.

9. True or False: The correct answer to 'Which of the following best explains what happens when a new device is connected to the Internet?' is 'An Internet Protocol (IP) address is assigned to the device.'.

10. True or False: The correct answer to 'Of the following, which gives the best upper bound for the value of f(N) where f is a solution to the recurrence f(2N + 1) = f(2N) = f(N)...' is 'O(log N) + 0(1)'.

End of Paper